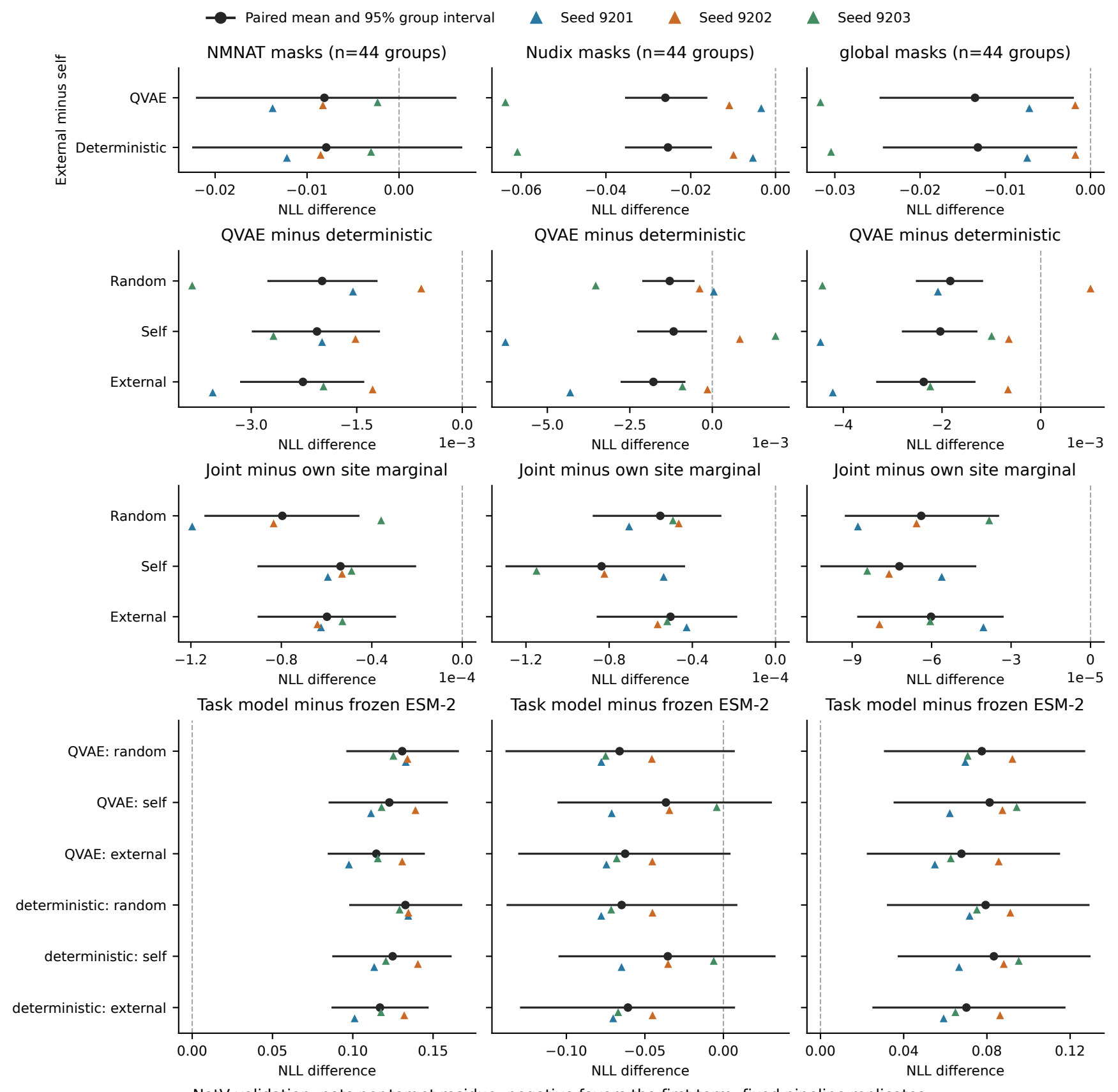
Paired mean and 95% group interval
Seed 9201
Seed 9202
Seed 9203
External minus self
NMNAT masks (n=44 groups)
Nudix masks (n=44 groups)
global masks (n=44 groups)
QVAE
Deterministic
−0.02
−0.01
0.00
−0.06
−0.04
−0.02
0.00
−0.03
−0.02
−0.01
0.00
NLL difference
QVAE minus deterministic
Random
Self
External
−3.0
−1.5
0.0
−5.0
−2.5
0.0
−4
−2
0
1e−3
Joint minus own site marginal
−1.2
−0.8
−0.4
0.0
−9
−6
−3
1e−4
1e−5
Task model minus frozen ESM-2
QVAE: random
QVAE: self
QVAE: external
deterministic: random
deterministic: self
deterministic: external
0.00
0.05
0.10
0.15
−0.10
−0.05
0.00
0.04
0.08
0.12
NatV validation: nats per target residue; negative favors the first term; fixed pipeline replicates